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PALEOMAGNETISM OF THE LATE PENNSYLVANIAN AND PERMIAN: A TEST OF THE ROTATION OF THE COLORADO PLATEAU

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Abstract. Coeval formations on both the North American craton and the Colorado Plateau have been sampled for the comparison of paleomagnetic pole positions. The latest Pennsylvanian-earliest Permian Laborcita and the Early Permian Abo formations of eastern New Mexico, and the latest Pennsylvanian Wescogame Formation of the Supai Group of northwestern Arizona were studied. Thermal demagnetization demonstrated that the majority of the samples in all three formations were remarkably free of significant secondary overprinting. Computation of pole positions used only samples which exhibited univectorial decay to the origin over at least three or more temperature steps. The paleopole of the Wescogame Formation is 119.2°E, 47.3°N (N=45 samples, k=41.6, dp=1.7, and dm=3.4); the Laborcita Formation paeopole is 130.6°E, 41.8°N (N=114 samples, k=42.9, dp=1.0, and dm=2.0); and the overlying Abo Formation paleopole is 118.3°E, 49.7°N (N=84 samples, k=54.9, dp=1.1, and dm=2.1). Comparison of the Wescogame, Laborcita, Abo and other paleopoles of this age shows the plateau poles displaced clockwise, confirming a clockwise rotation of the Colorado Plateau with respect to stable North America. The new data could be interpreted to indicate a 7° ± 4° rotation of the plateau; however, the direction of displacement of Pennsylvanian and Permian paleopoles is significantly different from that exhibited by the Early and the Late Triassic poles. The current body of data is best fit by not one, but two, rotations and suggests that the Colorado Plateau rotated in Late Permian time, prior to a later, presumably Laramide, rotation. Separate calculations of the Late Permian rotation, from Late Pennsylvanian data and from Early Permian data, both indicate a rotation of about 9.5° ± 2.7° northward and 5.4° ± 2.7° westward. This study thus suggests that at least two rotations of the Colorado Plateau have occurred prior to the most recent rotation involving the opening of the Rio Grande Rift. This history of rotations implies a quasi-independent plate motion behavior for the Colorado Plateau microplate.

Introduction

Detailed inspection of the paleomagnetic record recently showed that the Colorado Plateau has rotated relative to cratonic North America [Steiner, 1986; Bryan and Gordon, 1986]. One investigation used direct comparison of nearly coeval paleopoles [Steiner, 1986]. The other performed a maximum likelihood estimate of the effect of hypothetical rotations [Bryan and Gordon, 1986], using all available data and testing against a model of Carboniferous through Mesozoic apparent polar wander (APW) for North America [Gordon et al., 1984]. Significantly different magnitudes of rotation were obtained by the two methods. Steiner's [1986] method yielded 11° (± 4°), while Bryan and Gordon [1986] estimated 3.9° (± 2.6°).

The effect of the rotation of the Colorado Plateau was to translate Late Paleozoic and Mesozoic paleopoles along the same path as that of the APW of those times. The rotation and concomitant displacement of paleopoles is small relative to the precision of the paleomagnetic method. Because of these factors, analysis of the amount of rotation is highly sensitive to any sources of error.

To obtain an accurate estimate, only the best defined paleopoles for any time can be used; Bryan and Gordon's [1986] analysis included a number of data points that are of lesser or questionable reliability. Their analysis also is dependent on the validity of their APW model. The model assumes that APW paths are small circle segments and forces the data to fit the model. This has the inherent function of smoothing the data and hence diminishing discrepancies such as that arising from the rotation of the Colorado Plateau. Additionally, the model excludes the Late Jurassic cusp of the North American APW curve [Steiner, 1983; May and Butler, 1986]. Further, the Bryan and Gordon analysis assumes a pole of rotation for the plateau, one which does not fit the displacements documented between coeval paleopoles [Steiner, 1986].

Because of the smoothing effects produced by modeling and APW, comparison of paleopoles of the same age is preferred as the method of assessing the amount and timing of the rotation of the Colorado Plateau. The study reported here seeks to enhance the data base of coeval on- and off-plateau paleopoles. In this manner, it was hoped to refine the estimate of the amount of rotation and also to investigate a discrepancy posed by the Permian data set [Steiner, 1986].

In the previous analysis of published paleomagnetic poles of similar age, coeval paleopoles from both on and off the plateau were available from only five time periods [Steiner, 1986]. Four of these indicated approximately 10° of clockwise rotation of the plateau. However, Permian poles from the plateau and the craton appeared to be statistically indistinguishable [Gordon et al., 1984; Steiner, 1984, 1986] and indicated no rotation of the plateau. One explanation of the Permian discrepancy was possibly the lack of a well defined paleopole for the craton. Much of the craton data is from Peterson and Nairn [1971], who undertook a study to define the limits of the Kiaman interval. In so doing, they produced a number of Permian paleopoles. However, because their study was not primarily to define pole position, many of the paleopoles were calculated from only a few samples per formation.

Other reliable Permian paleopoles were derived

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from the Rocky Mountain Front Range, a region deformed in Laramide time. These sites thus had the potential of being rotated; hence the reliability of the poles was unknown. The Permian problem therefore seemed to lay in the definition of the craton pole, and a better defined pole was sought from a formation equivalent in age to a plateau Permian formation. The Early Permian Cutler Formation of the Colorado Plateau had been studied [Gose and Helsley, 1972], and the Cutler Formation can be traced laterally into the Abo Formation of New Mexico [Baars, 1962]. The Abo Formation, cropping out east of the Rio Grande Rift and thus demonstrably on cratonic North America, was sampled for comparison with the existing Cutler data.

Latest Pennsylvanian paleopoles were a second source of uncertainty in the previous analysis. The uncertainty stemmed from the unusual easterly location of the plateau Honaker Trail Formation (HT) paleopole, raising questions regarding the origin of remanence in this formation [Scott, 1975a]. While the HT paleopole appears rotated clockwise relative to its cratonic counterparts, its pole position is statistically identical to that of the plateau Early Triassic formation. At the time of the study, Scott [1975a] had suspected remagnetization, and now we know that cratonic Late Pennsylvanian paleopoles are not identical in position to the cratonic Early Triassic poles. In addition to the uncertainty of the HT remanence, problems also exist in the cratonic Late Pennsylvanian paleopoles, being either inadequately demagnetized or derived from locations affected by Laramide tectonics. These points are discussed in detail later.

Investigation of two Pennsylvanian formations in close proximity, in which there was no possible complication from subsequent Laramide compression, offered an opportunity to evaluate the rotation of the Colorado Plateau. The latest Pennsylvanian - earliest Permian was a time of continuous sedimentation on many parts of the North American continent, and as a result, many formations span the boundary. The environment of deposition of many of the preserved sequences was marginal marine to continental, and consequently the faunal record is often poor. Since age definition is therefore somewhat imprecise, this time period is best considered as a unit, the latest Pennsylvanian-earliest Permian. Two formations of this age were sampled for paleopole comparison, the Laborcita Formation of New Mexico and the Wescogame Formation of the Supai Group on the Colorado Plateau.

Paleomagnetic Analysis

Supai Group, Wescogame Formation

Three nonoverlapping sections within the central portion of the Supai Group were sampled. The sections, located within 3.5 km of one another, represent successively younger portions of the sequence. The lowest section (PS3) is an exposure in the I-40 roadcut immediately west of Seligman, Arizona (113.0°W, 35.2°N). The higher two, nearly stratigraphically continuous sections (PS1, PS2) are roadcuts of U.S. Highway 66 at the bluff known as Chino Point, west of Seligman (Figure 1).

The upper two sections (PS1, PS2) span

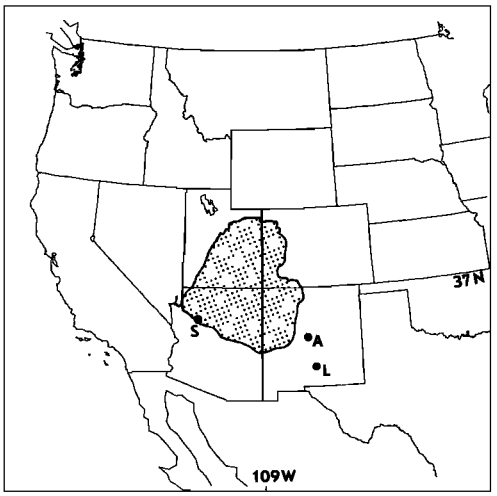


Fig. 1. Location of sampling sites; S, Supai Group, L, Laborcita Fm, and A, Abo Fm.

approximately the top one third of the Wescogame Formation, which is Virgilian in age [McKee, 1982]. The highest (PS1) encompasses 17 m of stratigraphic section, while PS2 encompasses 14 m. These two sequences are interbedded fine-grained red sandstone and siltstone. PS3 encompasses 17 m and is composed of calcareous, crossbedded, often massively bedded, pink and red fine-grained sandstone. The outcrop pattern in this area is highly discontinuous, and for that reason the exact stratigraphic postion of PS3 is unclear; it may be either lower Wescogame Formation or upper Manakacha Formation, both formations of the Supai Group. Its lithology most resembles that of the Manakacha [McKee, 1982; Elston and DiPaolo, 1979; Blakey, 1979; R.C. Blakey, personal communication, 1985; D.P. Elston, personal communication, 1985], which has an Atokan age [McKee, 1982]. The strata dip 1° to 7° to the northwest. A total of 165 samples were taken, all of which represent stratigraphically distinct horizons.

The natural remanent magnetization (NRM) of PS1 and PS2 samples form a streak from the present-axial dipole field direction to a shallow south-southeasterly reversed direction (Figures 2a and 2b). NRM directions of PS3 are much more scattered (Figure 2c), and the southeasterly magnetized samples have steeper inclinations than those of PS1-2.

Twenty-two pilot samples from the three sections were heated in 50° increments from 150° to 600° and to 630° and 645°C. (Subsequent to the completion of this study, it was recognized that parts of the furnace used in these experiments could be as much as 20°C low at temperatures above 400°C; therefore references to temperatures above this value should be read as such, for example, 600° = 580° to 600°C.) In the pilot demagnetization, much of the secndary Cenozoic magnetization is removed between 150° and 350°C. Many samples display stable, univectoral behavior above 350° to 450°C. The remainder of the samples were heated in 50°C increments from 350° to 600°C and to 630°C.

Figure 3 displays typical responses to demagnetization. Nearly one half of the samples

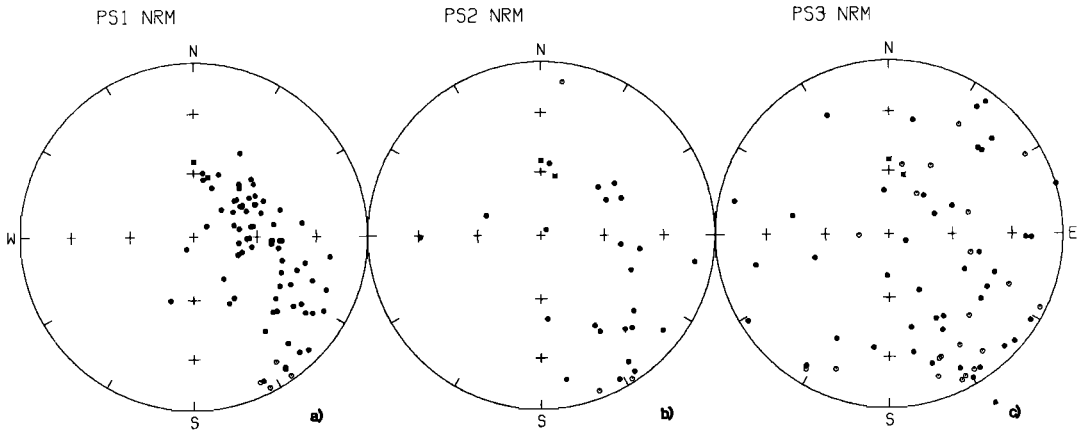


Fig. 2. NRM directions of the three Supai Group localities. Asterisks indicate the location of the present and axial dipole field directions. Equal area projection; open (closed) circles denote upper (lower) hemisphere directions.

from PS1 and PS2 displayed remanent magnetizations which were univectorial to the origin (Figure 3a); 22% of PS3 samples were similarly univectorial. The remainder of the sample populations retained secondary magnetizations, either to higher temperatures (>500°C) or throughout the demagnetization procedure (Figures 3b and 3c).

A minimum dispersion of directions is observed in PS1 and PS2 samples at 550°C and demagnetization behavior above that temperature usually deviates from linearity. This may suggest that magnetite carries some of the remanence in these samples, although the general remanence direction and intensity usually persist above 600°, indi-

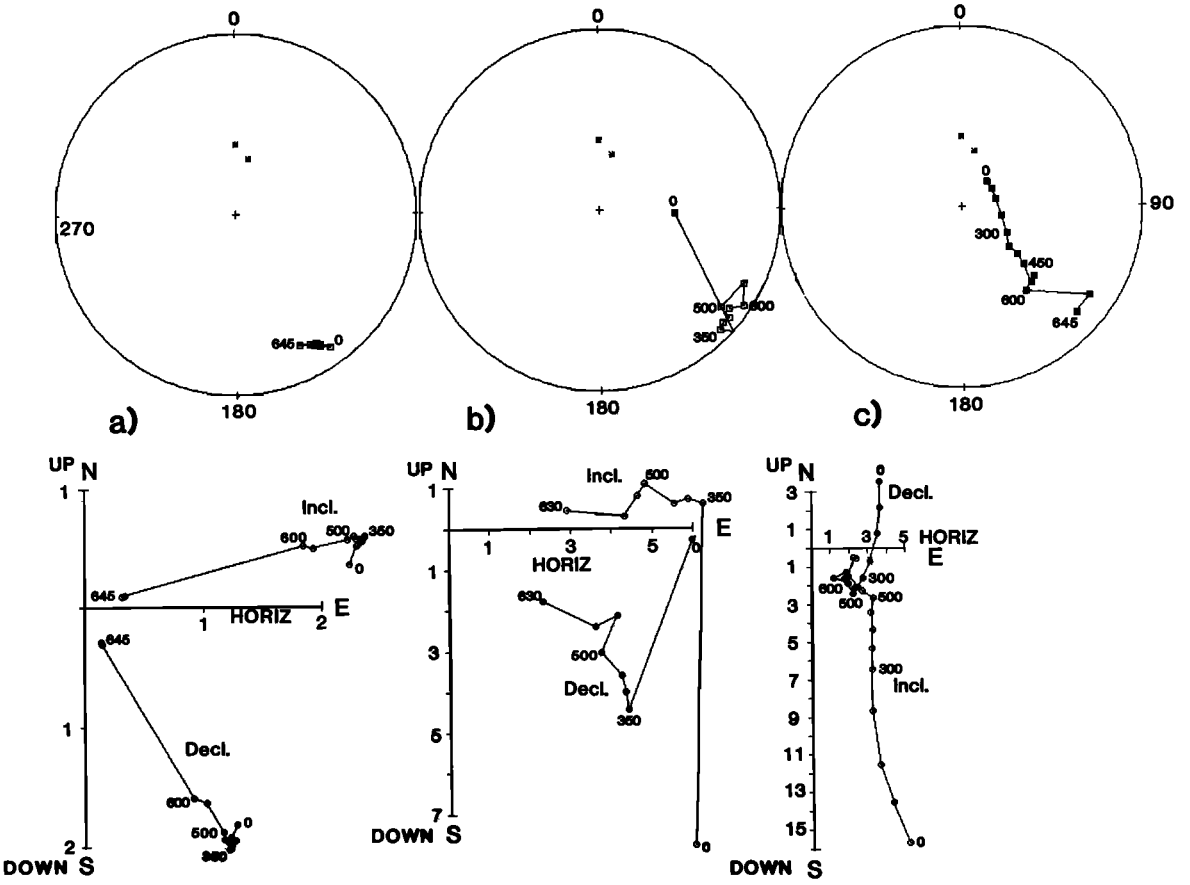


Fig. 3. Thermal demagnetization behavior of the Wescogame Fm. Stereographic projections are as in Figure 2. Orthogonal axes plots use the method in which inclination is plotted in the plane of declination. In these, the horizontal components of the magnetization (declination) are the solid circles. The vertical components, the angle between the vector and horizontal (inclination), are open circles; the distance from the origin is a measure of the total intensity of magnetization. Scale divisions: (a) 1 x 10^{-2} A/m (10^{-5}emu/cc), and (b) and (c) 1 x 10^{-4}A/m.

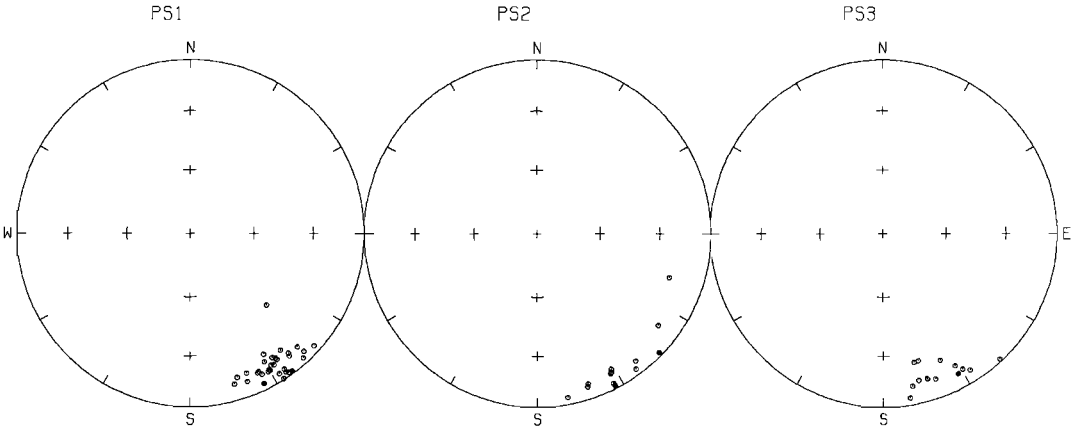


Fig. 4. Least squares directions of thermally demagnetized samples univectorial to the origin for the three Supai sites. Plots and symbols as in Figure 2.

cating a hematite carrier. The minimum dispersion of PS3 samples occurs at 600°C, however, with most samples showing continued progression of directions along demagnetization trends through 630°C.

Three-dimensional least squares line fits [Kirschvink, 1980] were computed for each sample to determine individual sample directions. Figure 4 displays the directions obtained from samples which exhibited univectorial decay statistically to the origin. Calculation of pole positions was conservative, using only samples having three or more demagnetization steps which displayed univectorial behavior to the origin. Samples such as that in Figure 3b were specifically excluded. Pole positions were calculated for each locality (Table 1 and Figure 5).

PS3 gives a significantly different pole position from that of the other two sites, and one which is in the wrong direction to be attributable to APW. The PS1 pole is calculated from 41% of the samples at that site, and the PS2 pole, from 50%; however, only 22% of PS3 samples fit the selection criteria. In addition to having markedly fewer samples univectorial to the origin and a significantly different mean direction-pole

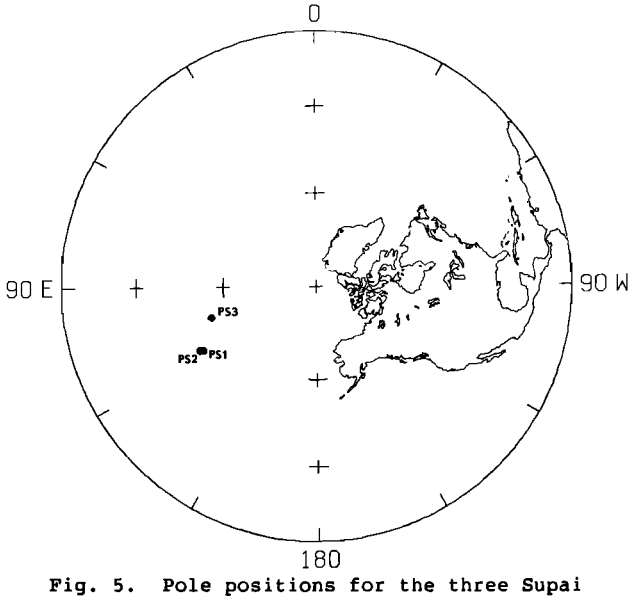


Fig. 5. Pole positions for the three Supai sites (Table 1).

position, PS3 also has NRM directions that were more dispersed and minimum dispersion of demagnetized directions occurring at a higher temperature. All of these features suggest either a greater degree of secondary overprinting or an additional period of overprinting with respect to the other two sites. The PS3 pole is also markedly similar to that of the Early Triassic and to that of the potentially remagnetized Honaker Trail Formation (discussed further later). All of these characteristics suggest possible remagnetization of this site, and suggest that the PS3 paleopole may be less reliable than the other two.

The exact age of the PS3 strata also is uncertain. For all of these reasons, the mean pole for the Wescogame Formation was calculated from only the PS1 and PS2 sites (Table 3).

Laborcita Formation

This formation is a marginal marine sequence of alternating limestone and fine-grained redbeds, of latest Pennsylvanian and earliest Early Permian age [Otte, 1959; Steiner, 1966; Steiner and Williams, 1967]. Two stratigraphically separated sections were sampled. A section encompassing 26 m at the base of the formation, in contact with the underlying Holder Formation, was sampled in La Luz Canyon, 9.5 km NE of Alamogordo, New Mexico (105.9°W, 33.0°N, Figure 1). A section through the central portion of this 170 m formation encompassed 33 m and was located in Fresnal Canyon, 2 km south of the lower section. The lithology of the sampled intervals is fine-grained red-brown siltstone. The geology of this area has been mapped and described in detail [Otte, 1959; Pray, 1961]. Strata dip 3° to the northeast. In the Fresnal Canyon section, a narrow (1.5 m) dike cuts the section. However, sampling avoided strata closer than 5.5 m either side of the dike, and no substantial effects were observed, as will be discussed. A total of 136 samples were collected, one half at each section.

The paleomagnetism of the Laborcita formation is unusually simple. NRM directions of both sections are largely contained in the SE quadrant (Figure 6) and form the distal end of a streak toward the present-axial field direction. The middle section is slightly more scattered, but most of this scatter is from a few samples which are within an interval of 5.5 to 6 m from the

TABLE 1. Supai Site Poles

Site	Dec	Inc	E Longitude	N Latitude	N	k	dp	dm
PS1	147.º	-10.2^{o}	119.4^{o}	47.6^{o}	32	58.2	1.7	3.4
PS2	147.3^{o}	-7.9^{o}	118.6^{o}	46.5^{o}	13	23.3	4.5	8.8
PS3	158.2^{o}	-11.9^{o}	106.5^{o}	54.6^{o}	14	38.5	3.4	6.7

N, number of samples; k, precision parameter; dp and dm semiaxes of ellipses of confidence.

dike, and also from a few samples of either green siltstone or coarser purple sandstone, lithologies not typical of the majority of the samples. Basically, most of the samples are remarkably free of significant secondary overprint.

Thermal demagnetization more quickly removed the secondary magnetization, in comparison to the demagnetization response of the Supai samples. Pilot samples were treated at the same steps as descibed for the Supai samples. By 250°C much of the secondary remanence has been eliminated (Figure 7a). The remainder of the samples were heated in 50° steps from 350° to 600° and to 630°C. Most samples are nearly univectorial to the origin above 350°C. A very few portions of the stratigraphic section and a few isolated samples have overlapping blocking temperature spectra, preventing isolation of the characteristic component (Figure 7b). However, in this formation demagnetization behavior univectorial to the origin is exhibited by all but 16% of the samples. The samples near the dike in the Fresnal Canyon section showed deviant NRM directions, but during demagnetization behave as the majority of the samples. They do not retain the secondary magnetization which gave rise to the aberrant NRM directions past the first heating step.

The characteristic direction of most of the samples is observed from 400°- 450°C to 550°-600°C. Figure 8 shows the directions obtained from least squares line fits for all samples univectorial to the origin. Calculation of mean directions and pole positions is the same as described for the Wescogame Formation. The two sections gave nearly identical mean directions and pole positions (Table 2). A formation pole position was calculated from all univectorial samples (84% of the population) and is shown in Table 2.

Abo Formation

Much of the Early Permian Abo Formation is exposed by a progression of roadcuts along U.S. Highway 60 (106.4^{o}W, 34.5^{o}N), near Abo Pass east of Scholle, New Mexico (Figure 1). Seven sites were collected, six in these roadcuts, and seventh from a natural exposure, as the top of the formation was only available from a hillside of low relief. These seven sites are the same as those used by Peterson and Nairn [1971] in their earlier study of the Abo Formation (their sites A1 through A8). The site locations span the entire 300 m of stratigraphic thickness of the Abo Formation, and sample about one third of it. Strata dip 2° to 6° to the northeast, except for crossbedded strata within channels in which the azimuths vary and dips are up to 20°. An additional site was collected slightly further south in a large roadcut on U.S. Highway 380 (106.3^{o}W, 33.9^{o}N), just east of Bingham, New Mexico (60 km south of the others). These beds dip 1° to the south-southwest.

Although Peterson and Nairn's [1971] earlier work was a carefully executed study, each site was represented by only 4 to 6 samples, and secondary magnetization clearly remained in many of the samples. The present collection was made to increase the number of samples used to calculate the Abo paleopole and to sample only the finer grained portions of the formation in order to eliminate secondary magnetizations from the pole

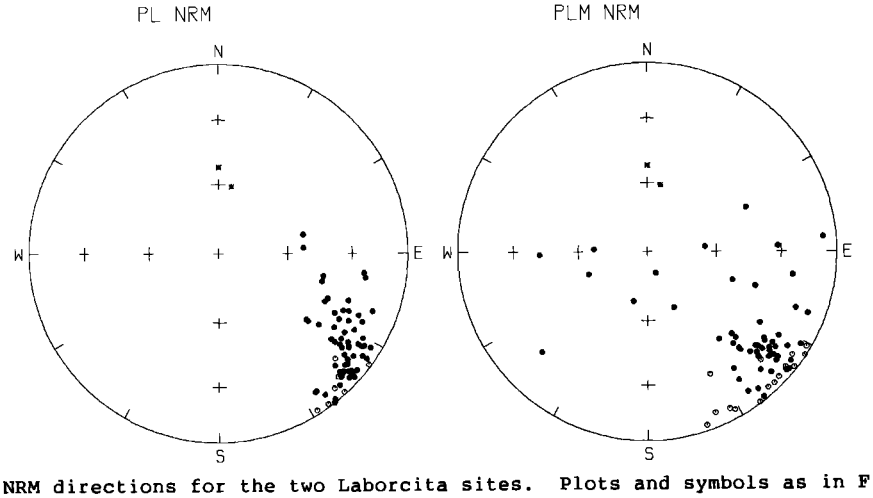


Fig. 6. NRM directions for the two Laborcita sites. Plots and symbols as in Figure 2.

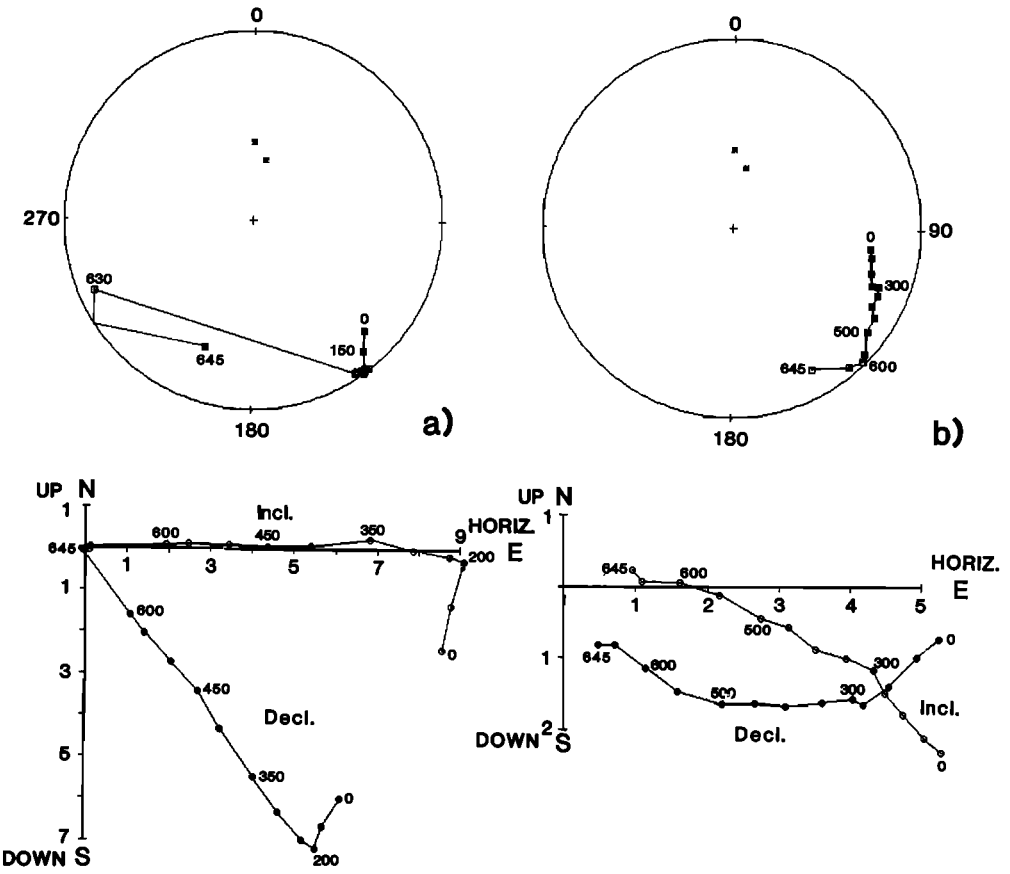


Fig. 7. Thermal demagnetization behavior of the Laborcita Fm. Plots are as in Figure 3. Scale divisions are 1 x 10^{-3} A/m.

calculation. A total of 129 samples were collected in normally bedded strata and another 23 were collected from crossbedded strata in the channels.

The Abo Formation has few diagnostic fossils, but its age can be clearly determined from its intertonguing relationship with the fossiliferous, fusulinid-bearing Hueco Limestone, as well as from its stratigraphic relationships to over- and underlying strata. Largely on the basis of the intertonguing relationship, the Abo is middle-late Wolfcampian and possibly extending into the early Leonardian [Pray, 1952; Otte, 1959].

The NRM directions of the Abo samples form a scattered streak of magnetic directions between a clearly defined SE negative direction and the presently observed geomagnetic field direction (Figure 9a). Most of the samples taken in the roadcuts display only small amounts of secondary magnetization (e.g., Figure 9b), while the single site from the natural outcrop has NRM directions that largely reflect secondary magnetization

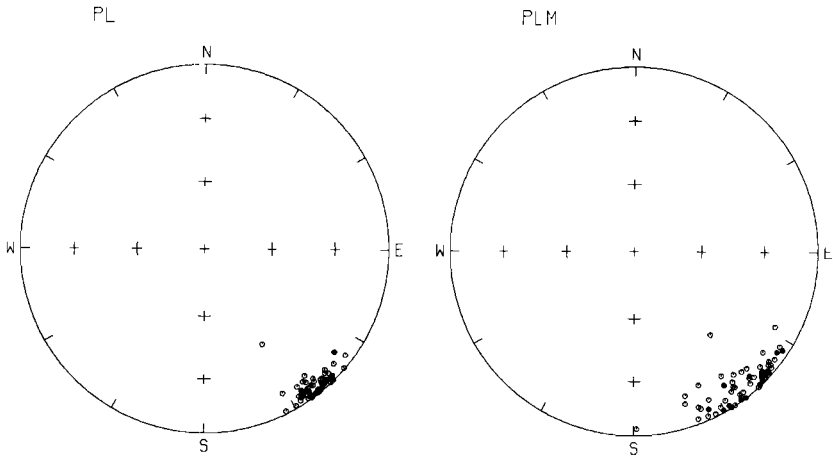


Fig. 8. Least squares directions of thermally demagnetized samples univectorial to the origin for the Laborcita sites. Plots and symbols as in Figure 2.

TABLE 2. Laborcita Site Poles

Site	Dec	Inc	E Longitude	N Latitude	N	k	dp	dm
PL	141.2°	-3.7°	131.8°	42.1°	55	85.7	1.0	2.1
PLM	141.0°	-4.9°	132.5°	42.4°	59	34.0	1.6	3.2

(Figure 9c) and account for the shallow northeast directions of Figure 9a.

Twenty-five pilot samples were stepwise thermally demagnetized, using the same temperature steps as the two previously discussed formations. As in the Laborcita Formation, most of the samples were univectorial to the origin (Figure 10a). Samples with substantial Cenozoic overprints generally display remanence components with overlapping blocking temperature ranges which can not be separated (Figure 10c). Minimum dispersion of directions occurs at both 500° and 550°C, with a slightly greater clustering, as expressed by the precision parameter k at 550°C. A number of samples exhibit abrupt intensity loss between 550° and 600°C, suggesting that magnetite carries some of the remanence of these strata; however, remanence directions often remain about the same and linear to the origin above 550°C, indicating a hematite carrier.

All of the samples from the natural outcrop fail to trend to the origin. However, 78% of the recent exposure (roadcut) samples were linear to the origin. Least squares lines were fit to all samples. Poles were computed separately from those such samples occurring within channels and for the vast majority from originally flat-bedded strata (Table 3). Samples from within the channels had a significantly higher dispersion than the other samples, although their pole position differs only slightly from the pole of originally flat lying strata. Both this greater dispersion and the indication of magnetite in the demagnetization behavior suggest a depositional remanent magnetization (DRM); the more scattered remanence directions presumably reflect either the effect of deposition on dipping surfaces or influence by currents or both. Despite the fact that their inclusion little changes the pole position, the channel samples are not included in the Abo paleopole considered most representative of this formation (Table 3). The univectorial, nonchannel samples used to calculate the pole position are shown in Figure 11a.

Two sites from this sample population dramatically illustrate the greater accuracy of paleomagnetic data that fresh exposures provide. As was mentioned earlier, the single site from the naturally exposed hillside had large secondary magnetization contributions in all samples (Figure 9c), and even with thermal demagnetization, it was not possible to isolate a characteristic magnetization in any of those samples (Figure 11c). In sharp contrast are the samples from the site in the largest of the roadcuts. This roadcut is about 20 m deep (others range from 3 to 10 m), and its samples had almost no secondary component (Figure 9b). All of these samples had univectorial decay to the origin which either included the NRM direction or began at low temperature. The site exhibits well grouped directions (Figure 11b), all of which were used in the paleopole calculation.

Comparison of Late Pennsylvanian and Early Permian Paleopoles

The pole positions of this study are comparable in age to a number of published paleomagnetic poles. Existing Late Pennsylvanian poles are those of the Honaker Trail Formation and the Halgaito Formation, both of the Colorado Plateau, the Dunkard Formation of the eastern United States, and part of the Casper Formation of southern Wyoming; Early Permian poles are those of the Cutler Formation of the Colorado Plateau, part of the Casper Formation, and the Ingleside Formation of northern Colorado. Three other Pennsylvanian pole positions, derived from sites in the eastern United States, are slightly older

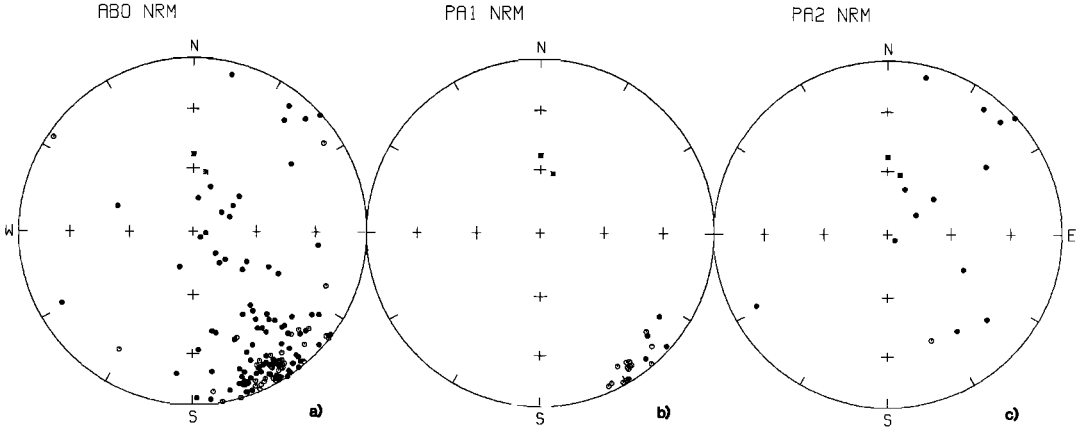


Fig. 9. NRM directions for (a) the entire Abo Fm. and (b) for a deep roadcut and (c) a natural outcrop. Plots and symbols as in Figure 2.

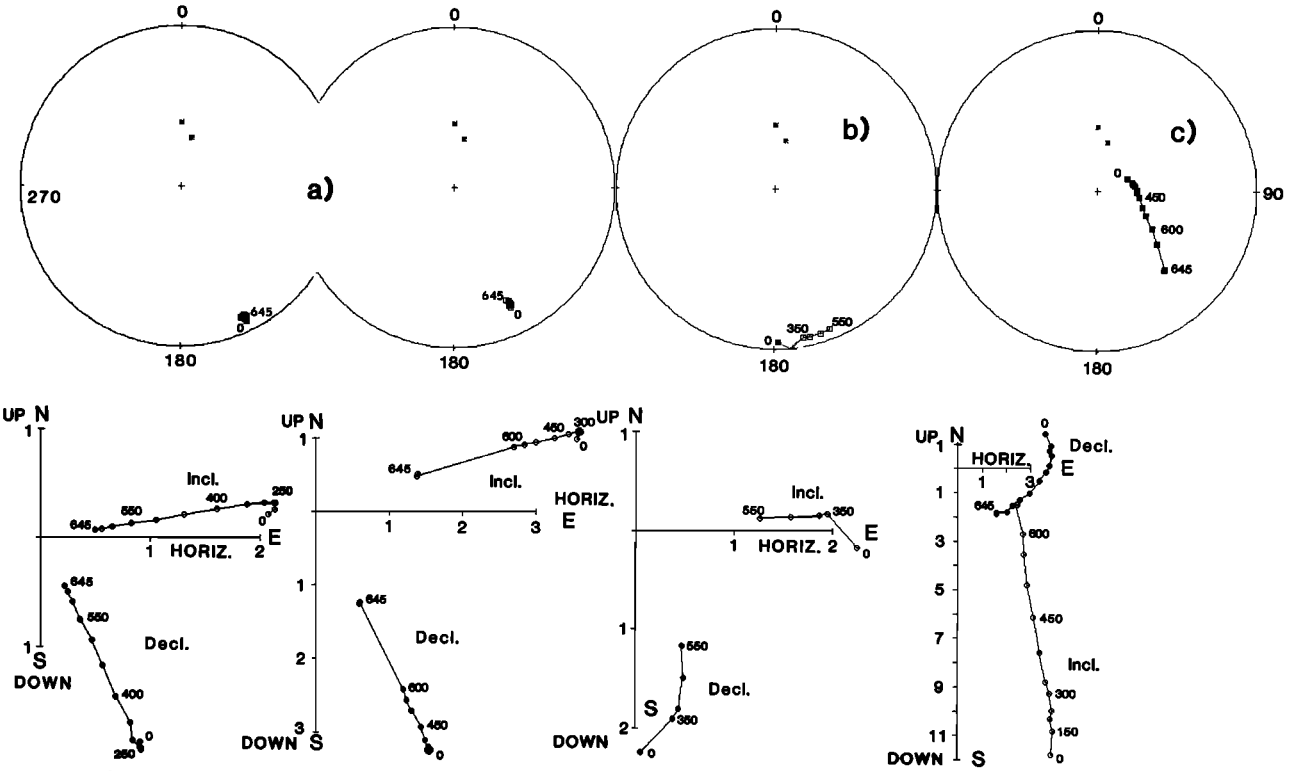


Fig. 10. Thermal demagnetization behavior of the Abo Fm. Plots as in Figure 3. Scale divisions are 1 x 10^{-2}A/m for Figures 10a and 10b and 1 x 10^{-3}A/m for Figure 10c.

(Middle-Late Pennsylvanian), but cannot be considered reliable indicators of the Late Carboniferous paleopole position, because in two instances of the short time sampled, and in the third, a lack of published results [Steiner, 1986]. Reevaluation of the published data in light of modern standards requires further rejection of some pole positions. The following discussion will evaluate the reliability of published poles. the comparableness of ages of the pole positions, and will compare the paleomagnetic results.

Latest Pennsylvanian-Earliest Permian

The Laborcita Formation of this study has a lithology transitional between the underlying marine Late Pennsylvanian (Virgilian stage) Holder Formation (predominantly limestone) and the overlying Early Permian nonmarine redbeds of the Abo Formation. The Laborcita intertongues with both formations, particularly in the area of sampling. The fauna is also gradational between latest Pennsylvanian and earlist Permian. The ammonites and gastropods are Late Pennsylvanian forms, while the brachiopods and fusulines were considered to be of Permian age [Otte, 1959]. The fusulinid fauna actually contains a mixture of Late Pennsylvanian and Early Permian taxa. Although the species of Permian affinity outnumber the Pennsylvanian ones, the Pennsylvanian species are found in more localities throughout the formation [Steiner, 1966]. These Permian fusulines are older than the Early Permian (Wolfcampian) standard section of North America (the west Texas Neal Ranch section of Ross, [1963]). This older aspect led Steiner and Williams [1967] to propose that the Laborcita fauna be added to the Early Permian, Wolfcampian standard section of North America. However, the mixture of latest Pennsylvanian and earliest Permian faunal elements and the close lithologic affinities with the formations above and below suggests that the Laborcita Formation spans the period boundary. As the Wolfcampian standard section rests on an unconformity [Ross, 1963], the Laborcita probably was deposited during the time of this unconformity. The transitional and intertonguing nature of both lithology and fauna suggests that the lower part of the Laborcita Formation represents time between the standardly defined Pennsylvanian and Permian periods. The lower and middle parts of the for-

TABLE 3. Abo Paleopoles

Bedding	Dec	Inc	E Longitude	N Latitude	N	k	dp	dm
Flat	152.7°	-5.8°	118.3°	49.7°	84	54.9	1.1	2.1
Cross	155.3°	-6.7°	115.5°	51.4°	17	21.0	4.0	8.0

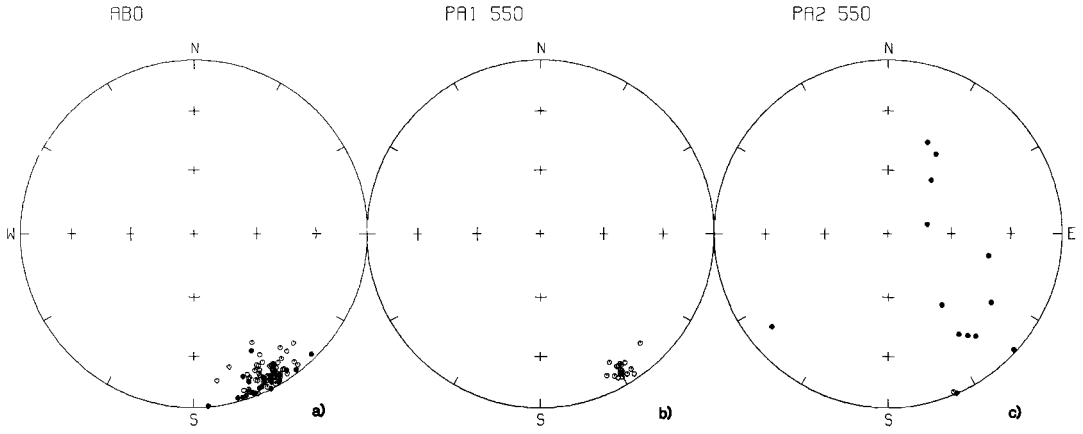


Fig. 11. (a) Least squares directions of thermally demagnetized samples univectorial to the origin for the Abo sites. (b) Directions of deep roadcut locality after 550°C. (c) Directions of natural outcrop after 550°C. Plots and symbols as in Figure 2.

mation were sampled for the pole position; the age of the pole therefore is probably latest Pennsylvanian and earliest Early Permian.

The Wescogame Formation (W) of this study is Virgilian [McKee, 1982], the youngest stage of the Pennsylvanian period. It is bounded by unconformities [McKee, 1982; Blakey, 1979]. The Laborcita Formation likely represents the time of the unconformity at the top of the Wescogame Formation, and thus the two formations are probably closely sequential in time. As the faunal evidence indicative of the latest Pennsylvanian age for the Wescogame Formation is derived from fossils identified only to the generic level [McKee, 1982], it is possible that the two formations may slightly overlap in time. Although more precise correlation is not possible at this time, the ages are clearly close.

The Casper Formation spans the Middle Pennsylvanian Desmoinian (represented by only a few paleomagnetic samples) through the latest Pennsylvanian Virgilian stages and the early part of the Early Permian Wolfcampian stage. Two paleopoles were computed for the formation, one each for the Pennsylvanian (CA1) and the Permian (CA2) portions [Diehl and Shive, 1981]. The paleopoles are well-defined by a carefully selected data set of several hundred samples; 233 samples (42%) were used to calculate the poles. Other than the location of the site in the Laramide deformation zone of the Rocky Mountain Front Range, the Casper poles appear to be reliable pole positions for these times.

The Dunkard Formation has formerly been considered to be an accurate representation of the geomagnetic field for the craton, first of the Early Permian, and subsequently of the Late Pennsylvanian, the age change resulting from a study of the flora of the formation [Clendening, 1974]. However, the Dunkard samples studied paleomagnetically were mostly undemagnetized, although one quarter were demagnetized in alternating fields (AF). Two computations of the paleopole were presented, one from NRM data and one from the one quarter of AF demagnetized samples. The demagnetized pole is computed from only 22 samples. Moreover, AF demagnetization is inadequate as a means to isolate the characteristic remanence of redbeds. Therefore neither the NRM nor the "demagnetized" pole is acceptable, and hence the Dunkard paleopole also can no longer be regarded as a completely reliable craton paleopole.

The Halgaito Formation paleopole [Farrell and May, 1969] was computed from only 14 samples which were demagnetized only once at 500°C. Both the number of samples and their treatment are inadequate to determine a reliable paleomagnetic pole position, and this pole is not considered further here.

The Honaker Trail (HT) Formation belongs to the Desmoinian, Missourian, and Virgilian stages of the Late Pennsylvanian [see Scott, 1975a]. Since HT and W both are located on the plateau, their pole positions should be quite similar; they are not (Table 4). This significant difference lends further weight to the suspicion that the HT Formation is remagnetized [Scott, 1975a]. Separate plateau and craton APW paths are constructed slightly later in this discussion (Figure 13), which show that the W pole occupies the same position in a plateau APW path as do Late Pennsylvanian poles in a craton APW path. Because of the coincidence of the HT pole with the Early Triassic paleopole position and the marked difference with respect to other Late Pennsylvanian paleopoles, Scott [1975a] had hypothesized a postdepositional, diagenetic magnetization. Diehl [1977] reached the same conclusion from the poor comparison of HT with the coeval Casper Formation paleopole and the statistical similarity to the Early Triassic, although the rotation of the plateau now makes it inappropriate to compare HT and Casper paleopoles. Nevertheless, the location of the coeval W pole in the correct position in the APW curve while the HT pole occupies the Early Triassic position is further circumstantial evidence that the Honaker Trail Formation has been remagnetized.

Other data sets also suggest a similar remagnetization episode. One is the Supai Group pole of the present study, PS3, which is practically identical to the HT pole (Tables 1 and 4), and to the Early Triassic pole position. A number of remanence characteristics suggested an inferior remanence record in PS3 relative to the other two sites and remagnetization of PS3 as the most plausible explanation for the differences between them. The similarity with HT and the Early

TABLE 4. Mean Poles

Formation	E Longitude	N Latitude	N	k	Alpha 95	dp	dm
Wescogame	119.2°	47.3°	45	41.6	---	1.7	3.4
Laborcita	132.1°	42.3°	114	48.2	---	1.0	1.9
Abo	118.3°	49.7°	8	100.5	1.6	---	---
Honaker Trail	106.4°	55.0°	41		1.5		
Cutler Rev.	110.7°	50.5°	66		2.3		
(Abo of Peterson and Nairn [1971]	116.0°	46.0°	6 sites	77.0	8.0)		
Craton L. Pennsylvanian (CA1 and L)	130.6°	44.1°	2 formations	746.4	9.2		
Craton E. Permian (CA2, I and A)	121.3°	48.8°	3 formations	712.8	4.6		

Triassic pole position further argues for remagnetization.

Yet another study of the Supai Group [Peterson and Nairn, 1971] also displayed a remanence similar to that of the Triassic rather than the Pennsylvanian or Permian. The stratigraphic levels of the Supai Group sampled in this study are unclear from the publication, hence the age of the portion sampled is unknown. Nevertheless, the non-Pennsylvanian or Permian paleopole is striking.

Thus three Carboniferous studies on the plateau have yielded apparent Triassic remanence directions. The position of the poles seems to suggest that remagnetization occurred in the Early Triassic, however, as no plateau Late Permian poles exist, the period of remagnetization could have been in the Late Permian. While the suggestion of remagnetization is only circumstantial, the number of such results certainly weighs in favor of a period of remagnetization on the Colorado Plateau in the Late Permian-Early Triassic, a topic to which we will return later.

Early Permian

The Early Permian subsequent to the latest Pennsylvanian-earliest Permian time is represented by four paleopoles. The Abo Formation pole of this study (A) replaces that of Peterson and Nairn [1971]. This pole is similar to that obtained by Peterson and Nairn [1971], but is displaced from it in a direction consistent with the elimination of a small contribution from the present field direction (Table 4).

The one Early Permian Colorado Plateau paleopole, that of the Cutler Formation [Gose and Helsley, 1972], is coeval with "A". The Cutler pole, along with the underlying, intertonguing earliest Permian Elephant Canyon paleopole, has been recalculated by Helsley and Scott since the original study [Scott, 1975a, b; Diehl, 1977]. The recalculation used a technique [Helsley, 1973] to eliminate samples with appreciable secondary contribution of the present field [Scott, 1975a, b]. The result was a single pole (CU) quoted in these sources, although the study was never published.

The recalculation significantly changed the position of the Cutler pole, but also utilized a severely reduced sample set: 66 versus the original 349 samples. Because the details are unavailable, the stratigraphic distribution of these 66 samples and therefore the amount of Cutler time that they represent, is unknown. On close examination, the change in the pole position is not clearly the result soley of removal of a present field overprint. Therefore the representative nature of this pole for Early Permian of the plateau now appears somewhat uncertain.

Other craton Permian poles are those of the Permian portion of the Casper Formation (CA1) and the time equivalent Ingleside Formation (I) located further south along the Front Range [Diehl and Shive, 1979]. Both of these appear to be reliable indicators of the pole for this time. The location of the sampling sites of these poles in an area affected by Laramide tectonics had raised the possibility of rotation of these sites. But the CA2 and I poles are statistically the same as A. The Pennsylvanian Casper (CA1) and the Laborcita poles are also statistically identical. This coincidence of New Mexico and Wyoming-Colorado poles suggest that no significant rotation of the sites occurred as a result of Laramide tectonism.

Pennsylvanian-Permian APW

The poles of this study and their confidence limits are plotted in Figure 12a. All reliable Late Pennsylvanian and Early Permian poles are shown in Figure 12b. This time period was obviously a time of a small amount of APW, as each of three data sets which span the period boundary display a separation between the older and younger poles: APW is seen within the Casper Formation, between the superposed, intertonguing Laborcita-Abo formations, and between the Wescogame and Cutler formations (Figure 12b).

The paleomagnetic definition of the L pole is exceptional (see data discussion), and its age is well-defined. The relative positions of this pole and that of the overlying, intertonguing Abo Formation (A) appear to suggest that the APW occurred in earliest Permian, that is, at some time within the Wolfcampian stage of the Permian, and not between the Pennsylvanian and Permian periods.

Estimates of Plateau Rotation

Amount of Rotation. In Figure 12b it is clear that the plateau poles are displaced clockwise from the cratonic poles. However, the APW complicates estimation of the amount of the rotation of the Colorado Plateau from a comparison of individual pole positions. Ideally, the displacement

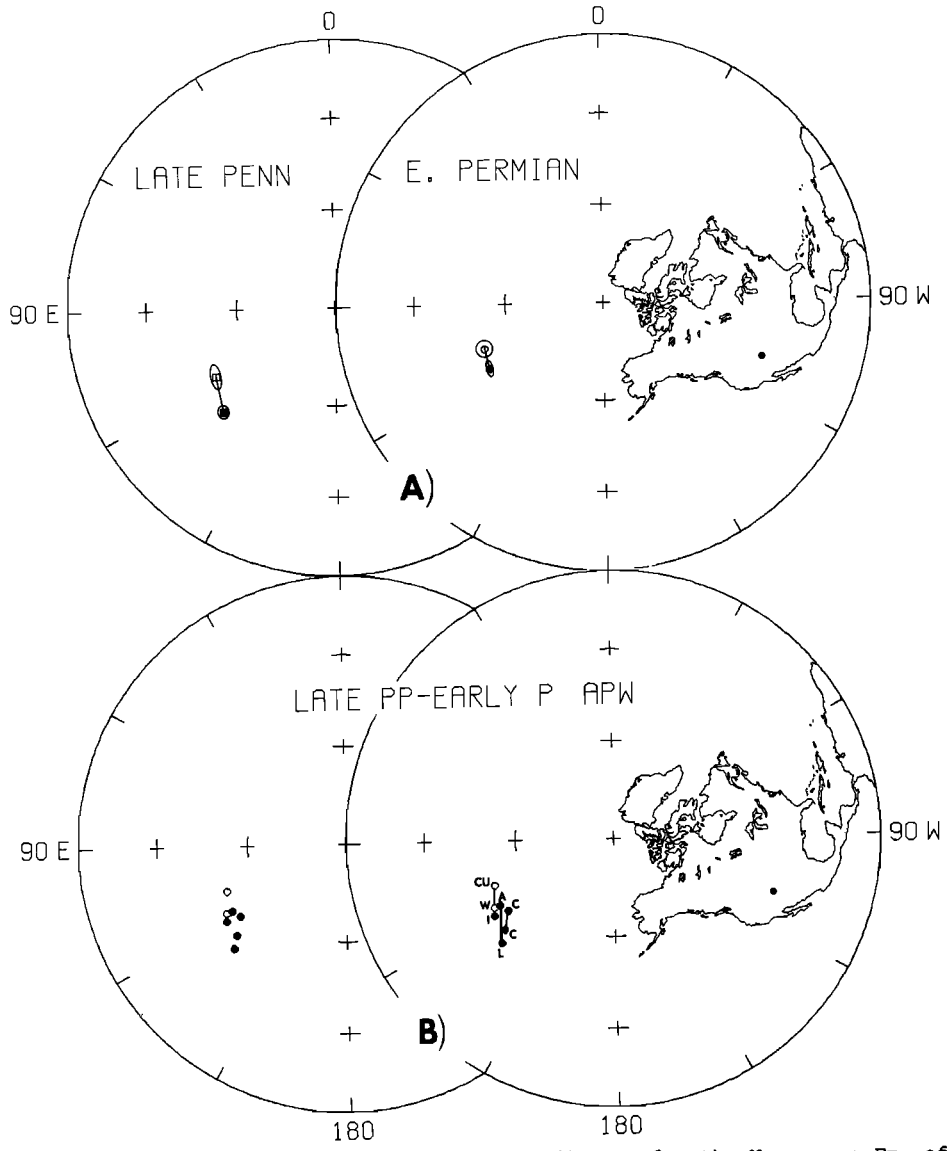


Fig. 12. (a) Left: paleopoles and ovals of confidence for the Wescogame Fm. of the Colorado Plateau (open circle) and the Laborcita Fm. of the craton (solid circle). Right: paleopoles and confidence limits for the Cutler Fm., Colorado Plateau (open) and the Abo Fm., craton (solid). Center of the Colorado Plateau is indicated by the black dot on western North America. (b) Reliable Late Pennsylvanian and Early Permian paleomagnetic pole positions for North America (solid) and the Colorado Plateau (open). Right plot illustrates APW during this time among three pairs of pole positions; the older has the more easterly longitude pole within each pair.

between those paleopoles which represent exactly the same time would determine the amount of rotation. However, among the Pennsylvanian poles, none represent exactly or solely the same time. The L pole is quite close in age to, but probably slightly younger than, the W pole. L is also apparently partly the same age as the younger part of the Casper Formation. Curiously, the L pole lies much closer to the Late Pennsylvanian portion of the Casper Formation (CA1) than to its Early Permian portion (CA2, Figure 12b), even lying on the older APW side of CA1. The relative ages would lead one to expect L to fall between CA1 and CA2, but close to CA2. The cause of this discrepancy is unclear, although it is probable that the faunal associations are not well enough known in the continental-marginal marine faunas of the Late Pennsylvanian-Early Permian to delineate age correlations this precisely.

Rotations were calculated using the method of Beck [1980], with error limit calculations as modified by Demarest [1983]. The rotation is calculated with respect to the center of the Colorado Plateau (109.5°W, 37.5°N). The confidence limits on such rotation calculations are difficult to estimate accurately because the attempt to compare only coeval paleopoles severely restricts the number of paleopoles involved in the rotation calculation. Often mean poles have been calculated from only two or three formation poles. Fisher statistics on such small numbers (N) gives large alpha 95 confidence limits on the

TABLE 5. Rotation and Translation of the Colorado Plateau

	Rotation Eastward			Translation Northward		
Direct Comparison of Late Pennsylvanian-Early Permian Poles						
Plateau Wescogame - Craton L. Pennsylvanian	8.2°	+3.8°	(9.9°)	2.7°	+3.8°	(9.6°)
Plateau Cutler - Craton E Permian	6.6°	+3.1°	(5.2°)	2.5°	+3.1°	(5.1)
Craton P/PP - Plateau P/PP	6.6°	+3.9°	(15.0°)	2.5°	+3.9°	(15.0°)
Inferred Late Permian Rotation						
Late Pennsylvanian poles	-4.9°	+2.7°	(13.1°)	9.9°	+2.6°	(12.7°)
Early Permian poles	-5.8°	+2.8°	(6.4°)	9.0°	+2.8°	(6.4°)

Rotations refer to the center of the Colorado Plateau. The first confidence limits given reflect the largest alpha 95s among any of the poles contributing to a given mean pole; the values in parentheses are the alpha 95s generated in the computation (using the methods of Beck [1980] and Demarest [1983]) of the mean poles.

rotation, even when the confidence limits on the individual poles are quite small. This situation therefore can make some rotations appear statistically insignificant (as in the Chinle and Devonian rotations calculated by Steiner [1986]). To compensate for this problem, two estimates are given for the confidence limits of each rotation: (1) the often unduly large calculated alpha 95 and (2) the largest alpha 95 among the group of poles which had been averaged [May and Butler, 1986].

Three comparisons of pole positions were performed to determine the amount of the rotation of the plateau: comparison between the Late Pennsylvanian paleopoles, between the Early Permian poles, and between combined plateau and craton poles of these time periods. The L pole was used in the Late Pennsylvanian comparisons because of its markedly greater similarity to the Late Pennsylvanian part of the Casper Formation and the probability that much of the sampled interval has a latest Pennsylvanian age. A cratonic Late Pennsylvanian paleopole was calculated from L and CA1 and is listed in Table 4. An average cratonic Early Permian paleopole was calculated from A, CA2, and I, also listed in Table 4.

Comparison of the Late Pennsylvanian craton pole with the plateau W pole indicates a clockwise rotation of the Colorado Plateau of 8.2° ± 3.8°; comparison of the Early Permian craton pole with the Cutler pole implies a rotation of the plateau of 6.6° ± 5.1° (Table 5). The uncertainty generated by APW coupled with inexact ages makes it best to estimate the rotation by comparing all Late Pennsylvanian and Early Permian data for the craton with the equivalent plateau data. This comparison indicates a rotation of 6.6° ± 3.9° (Table 5).

The Late Pennsylvanian trend of displacement between plateau and craton poles is notably the same as that of the Early Permian paleopoles. All three calculations from Late Pennsylvanian-Early Permian paleopoles indicate a rotation of the Colorado Plateau of a magnitude of about 7° (±4°). The uncertainties associated with the plateau Cutler paleopole make its reliability somewhat unclear, hence the Early Permian estimate of plateau rotation is less well-defined than that provided by the Late Pennsylvanian data. The Cutler or another Early Permian formation clearly needs to be restudied and a paleopole derived from particularly clean material.

Dual Rotation. The new Pennsylvanian-Permian data now make it possible to construct separate plateau and craton apparent polar wander curves for Late Pennsylvanian through Late Triassic time (Figure 13). Four mean poles for each block are shown. The amount of APW between each of the four time periods is roughly the same in each curve. This suggests not only that these poles are representative of their respective times, but that the indicated APW is approximately correct.

However, a significant feature is apparent from this diagram: the direction of displacement be-

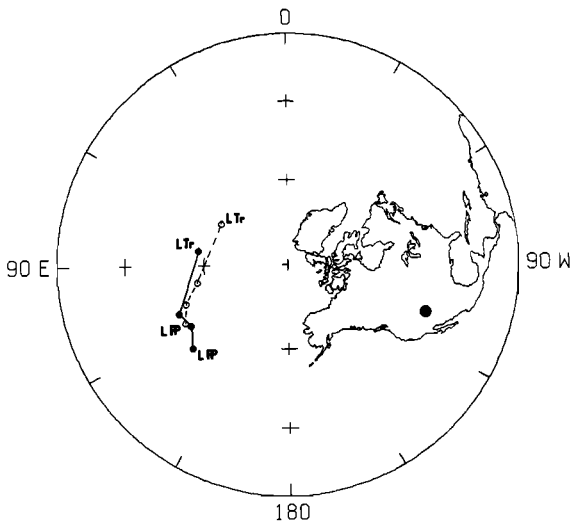


Fig. 13. Late Pennsylvanian through Late Triassic APW for the Colorado Plateau (open symbols) and the craton (solid symbols). The Early and Late Traiassic are from Steiner [1986]; the plateau Permian and Pennsylvanian are represented by the Supai fm. of this study and the revised Cutler Fm.; the craton Pennsylvanian and Permian are listed in Table 4. Center of the Colorado Plateau is shown as a large solid dot.

tween the Pennsylvanian-Permian paleopoles differs noticeably from that of the Early and Late Triassic paleopoles. Figure 14 clearly illustrates this difference. As shown in both figures, the data strongly indicate not one, but two separate rotations of the Colorado Plateau.

The time of the earlier rotation is constrained by the age of the Permian and Early Triassic paleopoles. The Permian poles represent time through sometime in the late Early Permian (Leonardian), while the Early Triassic poles probably represent time from near, but not precisely at, the Permo-Triassic boundary. These ages constrain the earlier of the two rotations to have occurred within the Late Permian.

The simplest scenario which accounts for the paleomagnetic data of Figure 14 is that of a dual rotation, as depicted in Figure 15. First, rotation in the Late Permian translated the Pennsylvanian-Permian paleopoles as shown in rotation 1 of the figure. The mean cratonic poles for the Late Pennsylvanian and the Early Permian (solid diamond and star, Figure 15) were projected to a position (open diamond and star) whereby the succeeding rotation of the plateau would displace them in the same direction and magnitude as the displacement between the Early and Late Triassic poles. This analysis suggests the following amounts of Late Permian rotation with respect to the center of the plateau (lower portion of Table 5): Late Pennsylvanian poles imply a displacement of the plateau by 9.9° northward and 5° westward, while the Early Permian poles give 9.0° northward, 6° westward.

The Euler pole for the Late Permian rotation must lie along the great circle bisecting the Pennsylvanian-Permian pole pairs. The near parallelism of the two pole pairs suggests that the Euler pole was probably distant from the paleopoles, perhaps 90° away. An Euler pole 70° to 90° to the south of the paleopoles along the great circle would be the Euler pole closest to the plateau itself (and would lie slightly east of the present day position of the Cook Islands). It is more reasonable to expect that a pole describing the rotation of a block would lie closer rather that farther from that block, and in a tectonic sense, plate interactions to the south of the plateau (Pacific-Caribbean-South America area) are likely to have been the cause of movement of the plateau. Hence the Euler pole is tentatively regarded to be in this area.

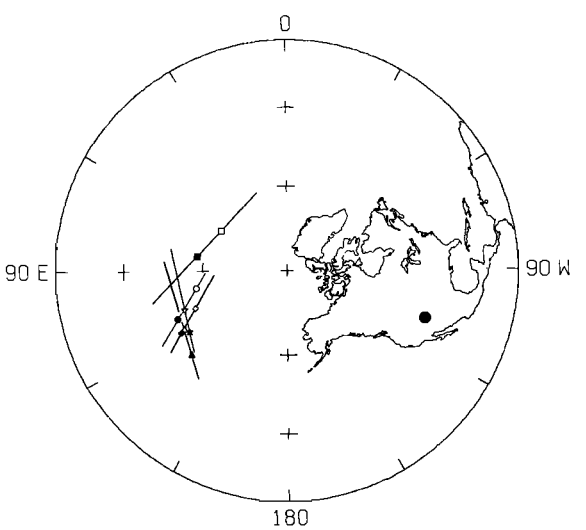


Fig. 14. Displacement between plateau (open) and craton (solid) paleopoles. Paleopole pairs are Late Triassic (squares), Early Triassic (circles), Early Permian (stars), Late Pennsylvanian (triangles), and Middle-Late Devonian (diamonds). Center of the Colorado Plateau is indicated by a large solid dot.

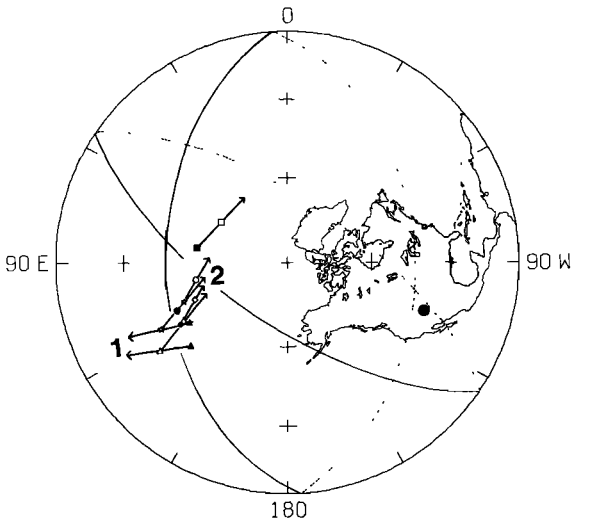


Fig. 15. The two inferred rotations of the Colorado Plateau. Arrows and numbers indicate rotations and sequence. Great circle loci of poles of rotation are shown for each rotation; dashed curves indicate southern hemisphere. Center of the Colorado Plateau is indicated by the large solid circle.

Continuing with the scenario of Figure 15, after the Permian rotation sedimentary accumulation resumed in the Early Triassic and continued until presumably Laramide time. A second rotation then occurred about an Euler pole along the second great circle of Figure 15, rotating the already displaced Pennsylvanian and Permian poles along with those of Triassic and younger times (displacement "2" in Figure 15).

This second rotation is best defined by the displacements of the Early and Late Triassic paleopoles calculated by Steiner [1986]. Again, the displacements of the paleopoles are nearly parallel, suggesting a distant Euler pole. The geologically most reasonable pole would again be one lying somewhat close to the plateau, and in this case an Euler pole 90° away from the paleopoles would lie in the Pacific Ocean, due west of southern Baja California (Figure 15).

The two rotations derived from the scenario of Figure 15 are of similar magnitude, the first producing a rotation of the center of the plateau of about 9.5° northward, while the second [see Steiner, 1986] rotated the plateau about 11° eastward.

Consequences and Implications

Tectonic activity necessary to induce a rotation of the Colorado Plateau in the Late Permian is compatible with the absence of Late Permian sedimentary strata on the plateau. Late Permian strata do occur on the craton, not far from the

edge of the Colorado Plateau (west Texas, eastern New Mexico). Tectonism must have occurred to either prevent deposition or to erode these strata and therefore their absence independently indicates tectonic activity involving the Colorado Plateau in the Late Permian.

Apparently contradictory to a rotation of the Colorado Plateau in the Late Permian is the displacement displayed by the Devonian paleopoles [Steiner, 1986]. As is shown in Figure 14, the rotation of the Devonian poles mimics those of the Triassic poles, and not that of the earlier Pennsylvanian-Permian poles. But the Devonian poles are known to be remagnetized poles. A strong case for Late Paleozoic, Kiaman, remagnetization has been presented by Irving and Strong [1984, 1985]. In their assessment, Irving and Strong noted that the Colorado Plateau declinations were rotated clockwise relative to cratonic Devonian declinations and concluded that the plateau had rotated subsequent to the remagnetization. The fact that the Devonian paleopoles lie close to those of the Permian and Triassic suggests that the remagnetization occurred in this time frame. By their position, the HT and Supai paleopoles also suggest remagnetization in this time period. Tectonism associated with the rotation of the plateau could be conducive to an episode of remagnetization. If the remagnetization were coincident with the Permian rotation of the Colorado Plateau, paleopoles of remagnetized formations would reflect only the second rotation of the Plateau, as the Devonian poles appear to do.

The Late Permian rotation of the Colorado Plateau is roughly coincident in time with the Sonoma Orogeny [Siberling, 1973; Speed, 1979]. However, the inferred northward and slight westward motion implies collision from the south-southeast. The event is probably related to a final closure of the pre-Gulf of Mexico area at the end of the Ouachita Orogeny [Dickinson, 1981; Pindell, 1985]. Although later in time, the counterclockwise motion inferred for the plateau is consistent with the left lateral interplate deformation which Budnik [1986] argues formed the Ancestral Rocky Mountains. Very recently, Handschy et al. [1987] have concluded that blocks of the Ouachita system of northern Mexico were thrust against North America in the Late Permian. Both the location of these blocks and the sense and timing of the thrusting is fully compatible with the Late Permian northwestward rotation of the Colorado Plateau suggested here on the basis of the paleomagnetic data.

Conclusions

New paleomagnetic pole positions have been obtained for the Late Pennsylvanian and Early Permian Wescogame, Laborcita, and Abo Formations. These new data demonstrate that coeval published paleopoles from sites within the Rocky Mountain Front Range have not experienced rotation as the result of Laramide tectonics. Abnormal positions of some of the Late Pennsylvanian poles suggest the possibility of a period of remagnetization in the Late Permian-Early Triassic.

Comparison of all reliable paleomagnetic data from the Late Pennsylvanian and Early Permian substantiates the earlier inference of rotation of the Colorado Plateau, but suggests that the Colorado Plateau has experienced, not one, but two rotations (prior to the opening of the Rio Grande Rift). Both rotations were of similar magnitude. The first rotation, about $9.5^{\circ} \pm 2.7^{\circ}$ northward, occurred in Late Permian time (end of the Paleozoic), while the second, about $11^{\circ} \pm 4^{\circ}$ eastward, is inferred to have happened in association with Laramide deformation (end of the Mesozoic). These two rotations and that associated with the Rio Grande Rift suggest that for some time the Colorado Plateau has behaved as a semi-independent micro-plate associated with the North American plate.

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